TWISTING BY THE PULLBACK OF THE TAUTOLOGICAL BUNDLE KILLS THE HIGHER DIRECT IMAGES ALONG THE PROJECTION OF A CLOSED SUBSPACE OF PROJECTIVE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

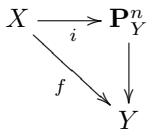
- Vanishing of a higher direct image can be checked etale locally on the target. Same chapter spaces-cohomology.tex, equation equation-representable-higher-direct-image (invoked at proof line 3828). It reduces the statement to the case where Y is the spectrum of a Noetherian ring.
- The scheme case of the vanishing after twisting. Cohomology of Schemes, label coherent-lemma-kill-by-twisting (invoked at proof line 3831). Once X is a scheme and the base is affine Noetherian, it supplies the integer d_0 past which the twisted higher direct images vanish.

Source reference. Original lemma 02O1: sources/stacks-chow/tags.txt, source lines 3500–3500. Navigation pointer for \ref{coherent-lemma-kill-by-twisting} used inside the counted span (line 3831); the argument that consumes it is retained in the span.

Source reference. Original equation 071Z: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 318–318. Navigation pointer for \ref{equation-representable-higher-direct-image} used inside the counted span (line 3828); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Consider a commutative diagram*



of algebraic spaces over S. Assume i is a closed immersion and Y Noetherian. Set $\mathcal{L} = i^*\mathcal{O}_{\mathbf{P}^n_Y}(1)$. Let $\mathcal{F}$ be a coherent module on X. Then there exists an integer d_0 such that for all $d \geq d_0$ we have $R^p f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) = 0$ for all $p > 0$.

Proof. Checking whether $R^p f_*(\mathcal{F} \otimes \mathcal{L}^{\otimes d})$ is zero can be done étale locally on Y, see Equation (071Z). Hence we may assume Y is the spectrum of a Noetherian ring. In this case X is a scheme and the result follows from Cohomology of Schemes, Lemma 02O1. □